Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 5

Cloud Storage

Requirements, block, file and object storage, and storage tiers.



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Recap

LAST SESSION

Compute service

Computing power delivered on demand over the internet.

LAST SESSION

AWS EC2

Amazon Elastic Compute Cloud: virtual machines on AWS.

Every one of those virtual machines needs somewhere to keep its data. That is today’s topic.

Overview today

1. What is cloud storage?
2. Requirements for storage
3. Why cloud storage?
4. Types of cloud storage: block, file and object
5. Object storage tiers

THIS WEEK

Cloud storage

From what any cloud storage must guarantee to choosing between block, file and object storage.

What is cloud storage?

Remote services that store data we reach *over the internet* or a private network connection.

We delegate the responsibility of data management, security, capacity planning and maintenance to a third party.

A cost-effective and scalable solution to store data and files, rather than managing the infrastructure on-premises.



PART 1

Requirements for storage

Storage is a fundamental component of any software application, so we must be able to access the data we store at all times.

Three fundamental requirements



Durability

Data will survive power failures, hardware failures, human errors and natural disasters.



Availability

Storing data in a cloud platform is useless if we cannot access it when we need it.



Security

Data must be secure, and we should be able to decide who gets access to it.

Moving data to remote locations is a critical decision. Any cloud-based storage solution must meet these three before we make it.

Durability

Durability measures the likelihood of data loss, as a percentage over a given year.

11	12	16
nines	**nines**	**nines**
99.999999999% for the Amazon S3 standard tier. Google Cloud storage services offer the same.	Durability offered by Microsoft Azure.	Offered by some Azure services.

With 11 nines and 1 billion objects, you can go 100 years without losing a single object.

How vendors achieve durability

Redundancy

Vendors make multiple copies of data. Replicating data in multiple places reduces the chance that all copies are corrupted or lost.

Three copies by default

On most cloud platforms the default is to make three copies of any data stored, often in the same datacenter.

Zones and regions

We can increase durability by spreading our data across zonal and regional boundaries.

Why it matters: durability is not fixed. We raise it by choosing where the copies of our data live.

Availability

Availability implies *uptime*.

The data we store anywhere is only useful if we can access it. Availability measures how readily available a service is.

Like durability, it is a percentage and is counted in a number of nines.

Availability	Downtime per year
99.5%	1.83 days
99.9%	8.76 hours
99.95%	4.38 hours
99.99%	52.56 minutes
99.999%	5.26 minutes
99.9999%	31.5 seconds
99.99999%	3.15 seconds

Availability by storage type

99.5%–99.99%

availability

The range many major cloud service providers ensure for their storage services.

Hot storage

Data we access often. Retrieval is fastest.

Archival storage

Retrieval takes longer than for data in hot storage, yet is still faster than any traditional way of data access.

Retrieval and access times vary with the type of storage we choose.

Security

We want our data secure at rest and in transit. Security is often the point where companies hesitate to adopt the cloud.

A matter of trust

Vendors invest heavily in the physical security of datacenters and in cyber security, and hold the certifications and accreditation to prove it.

Encryption

Cloud platforms encrypt data at rest and in transit. We also have the option to use our own keys.

Access control

A suite of tools helps us manage and decide who gets access to what, and for how long.

It is not far-fetched to say our data is more secure in the cloud than we could keep it in an on-premises environment.

Fully managed and accessible



Fully managed

Cloud vendors take care of security, durability, availability, recovery in case of failures and capacity planning behind the scenes. We simply store and access data on a fast, cost-effective, reliable service.



Accessible

Data is accessible from anywhere, at any time, so a distributed team can reach it through web browsers, mobile applications and desktop applications. An internet connection, or a private connection for a private cloud, is all that is required.

Why cloud storage?

01

Ease of use

A service exists for both traditional file-based and modern cloud-based storage.

02

Built-in security

Data is stored encrypted, and encrypted while in transit.

03

Scalable and performant

Fully managed and serverless: we keep adding data and the service keeps up.

04

Cost

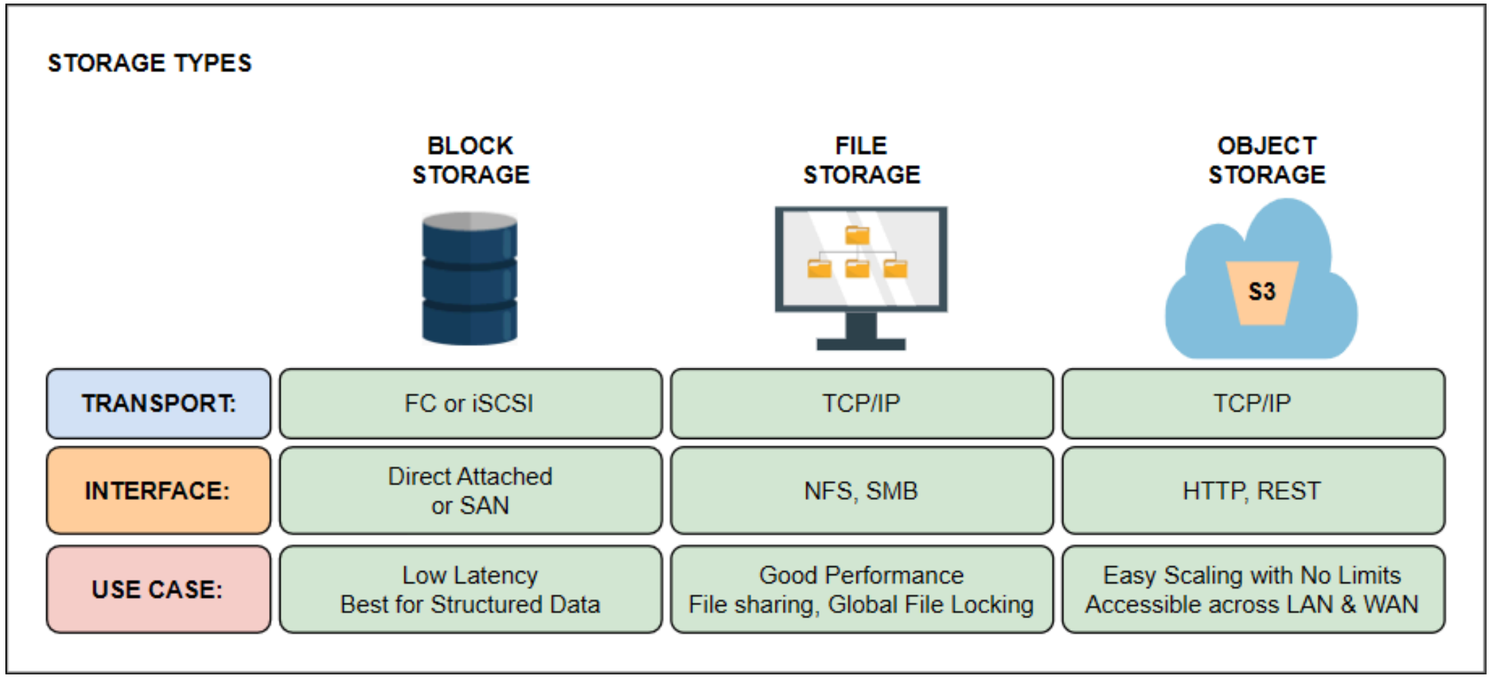
Among the cheapest services on any cloud platform.

05

Capacity

Almost infinite storage, like many other cloud services.

Types of cloud storage



There are three types of cloud storage: block, file and object.

STORAGE TYPE 1 OF 3

Block storage

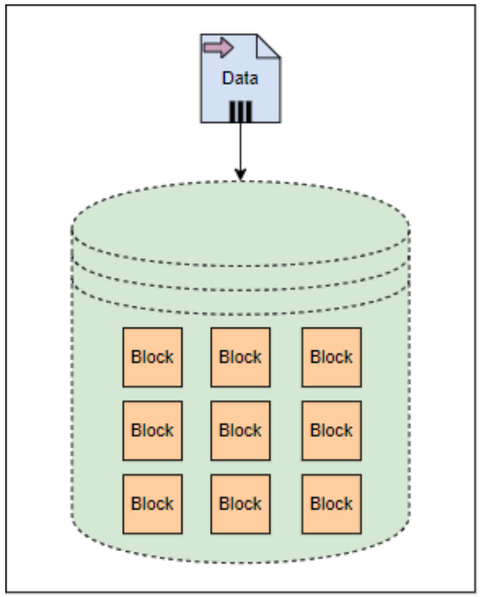
What is it? Where and when is it used? Which services do cloud platforms offer?

What is block storage?

Data is broken into *fixed-sized blocks* before it is stored.

The blocks are stored separately across many locations. Each block gets a unique identifier so it can easily be located.

Breaking data into smaller blocks allows for scalability, and stores data more efficiently than traditional file systems.



Data stored as fixed-size blocks

A building block of the cloud

Under the hood

The Infrastructure as a Service model, container orchestration and many other services use virtual machines, and virtual machines use block storage for data and the boot loader.

Attached as volumes

We cannot access block storage directly. Only the host operating system can, by attaching to volumes.

Transport

It uses Fiber Channel (FC) or Internet Small Computer Systems Interface (iSCSI).

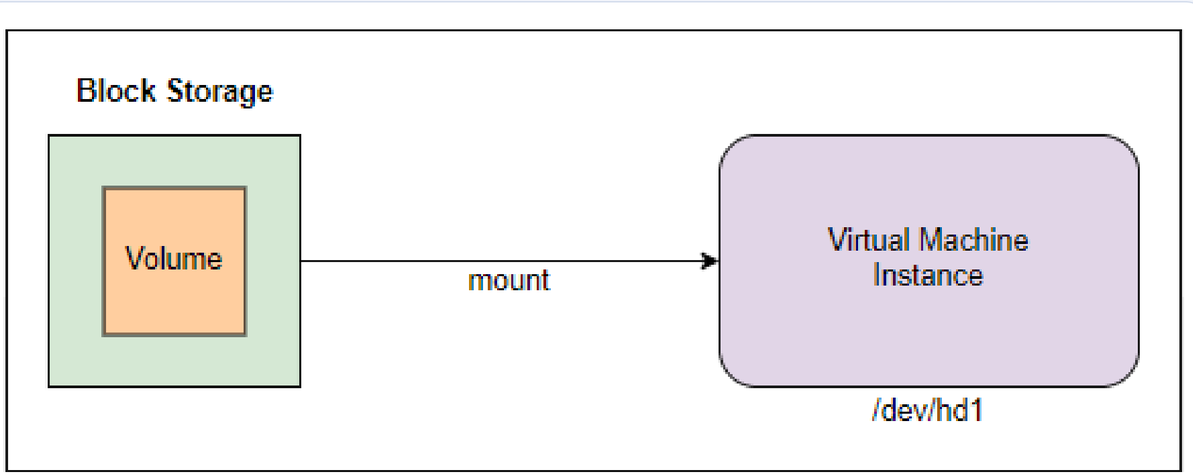
Why it matters: every virtual machine you launch, including an EC2 instance, boots from block storage.

Disks for virtual machines

Virtual machines use block storage for bootloaders, and as a volume for data storage.

A hypervisor or VM can mount multiple volumes for data storage.

The same volume cannot be attached to more than one virtual machine.



Mounting a block to a volume for storage

Where else block storage is used

Low latency applications

Ideal for applications that need low latency and high throughput. SQL Server, MongoDB and SAP use it for data persistence.

Backup and disaster recovery

Backups let us recover to the previous known good state. High durability and availability make block storage ideal for them.

Data migration

An efficient, cost-effective way to move data between environments, including from on-premises to the cloud.

Containerized workloads

Orchestrators such as Kubernetes run containers on VMs. A host VM can mount multiple blocks and use their speed.

Block storage services

Cloud platform	Service
AWS	Amazon Elastic Block Store (EBS) and Amazon EBS Snapshots
GCP	Google Cloud Persistent Disks
Azure	Azure Disk Storage

Each platform has its own name for the same idea. In this course we use the AWS one.

STORAGE TYPE 2 OF 3

File storage

What is it? How is it different from block storage? Where is it used?

What is file storage?

File-level storage in a *hierarchy* of folders and sub-folders.

It is normally associated with network attached storage (NAS) technology, and is similar to the local hard drives on our laptops and PCs.

The difference: a hard disk on our PC is accessible only to us, while file storage can be shared among many users.



The storage hierarchy in file storage

How file storage is accessed

Managed for us

A NAS (dedicated device) or network operating system (NOS) handles rights management, file sharing, file locking and other operations on files.

By a path

It is easy to configure, familiar to most people, and files can easily be accessed by a path.

File-level protocols

We use common protocols such as Network File System (NFS) or Server Message Block (SMB).

Unlike a block volume, which attaches to one virtual machine, a file share is used by many users at once.

Where file storage is used

File storage is a commonly used solution, since we all use it daily on our PCs, laptops and servers.

Collaboration

Good for unstructured data such as images, videos, text files and logs. Teams anywhere read and write to a common file share without learning a different tool.

Shared storage for containers

Containers are often stateless and keep no data after they shut down. File shares act as persistent volumes that multiple containers can access.

Web servers and content management

Web applications run on multiple servers that need the same set of files to serve content. File storage allows easy sharing.

File storage services

Cloud platform	Services
AWS	Amazon Elastic File System (EFS)
GCP	Google Cloud Filestore
Azure	Azure Files and Azure File Sync

Many cloud-based file storage solutions also provide desktop tools that act like local drives on PCs.

STORAGE TYPE 3 OF 3

Object storage

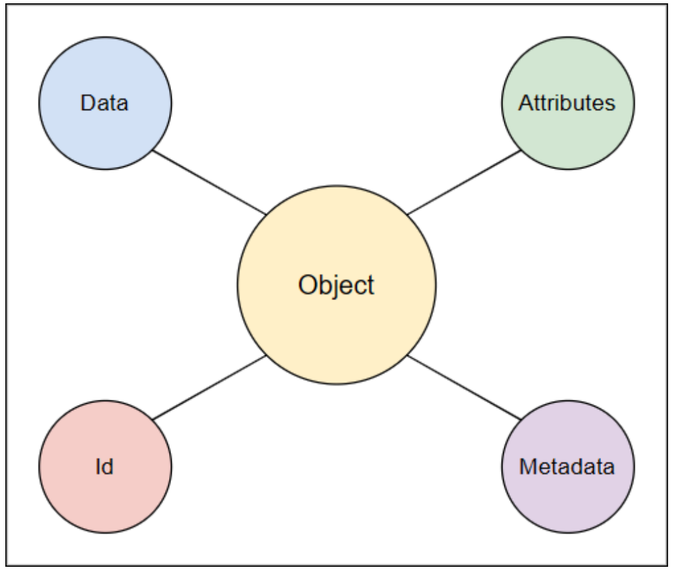
Often called the storage of the cloud, since it eliminates the scaling limits of traditional file systems.

What is object storage?

Data is broken into pieces called *objects*, kept in a container or bucket.

Each object has an identifier. Objects can then be distributed across multiple nodes or servers.

Unlike files in file storage, objects are stored in a flat structure. They do not follow any directory or folder.



The basic structure of an object

What an object contains



Identifier

Every object has a unique identifier.



Data

The content itself, for example a video, an image or an application log.



Metadata

Everything we need to know about the file: time of creation, size, subject. It is customizable, and helps us find objects quickly even among billions.



Attributes

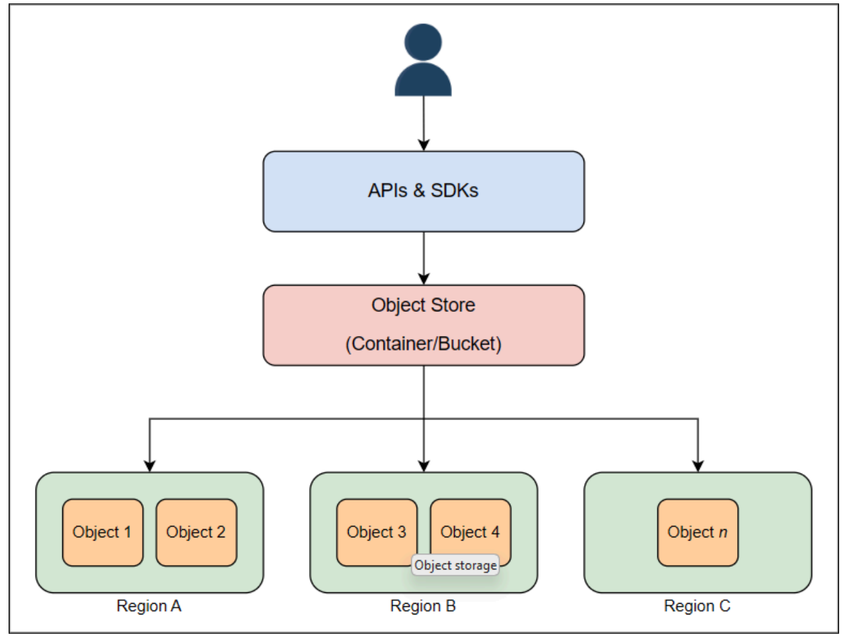
Help us define policies around the object, for example whether a user has the right to download, write or delete it.

How object storage works

We interact with the object store through APIs and SDKs, which abstract the complexity of how it works underneath.

Object storage is highly durable and highly available because objects are copied to different locations. By default, almost every major platform creates three replicas.

We can increase this by replicating objects across datacenters and regions.



Interacting with objects in object storage

Object storage services

Cloud platform	Services
AWS	Amazon Simple Storage Service (S3)
GCP	Google Cloud Storage
Azure	Azure Blob Storage

Cloud platforms have integrated object storage with many of the other services they offer.

Where object storage is used

01

Unstructured data

Images, audio, video, logs, and data for big data analysis and machine learning. No capacity planning.

02

Static data

Best for data that does not change often: write once, read many times.

03

Cloud-native apps

REST APIs and SDKs make it easy to build container, serverless and web applications.

04

Backup and recovery

A very cheap place for long-term data, VM snapshots and disaster recovery.

05

Archiving

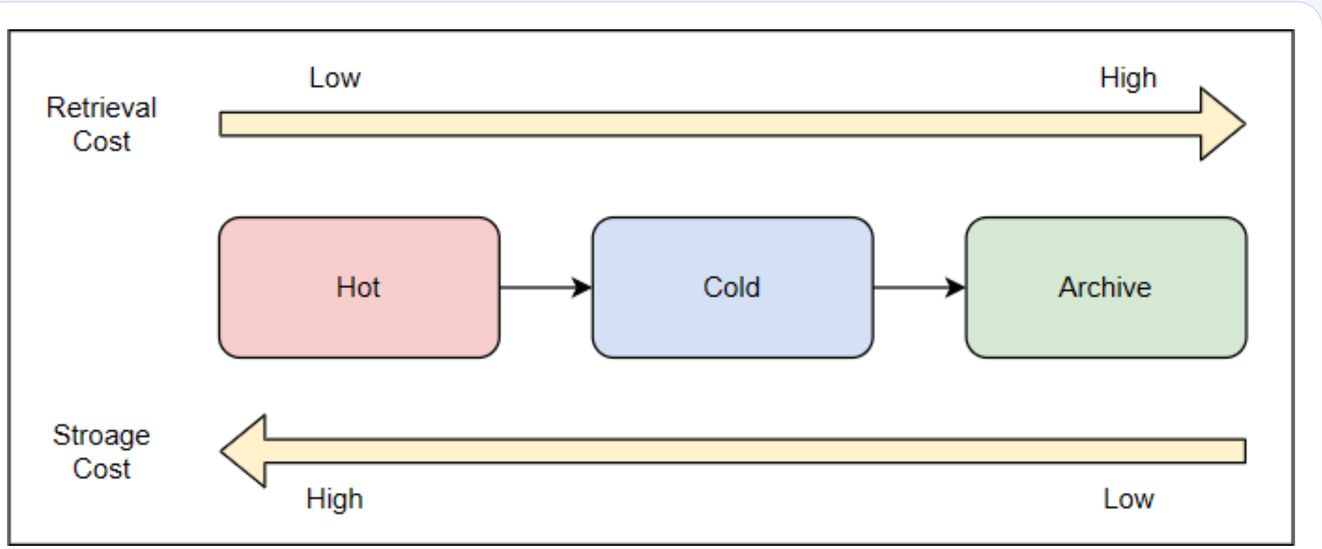
Storage tiers make long-term data retention cheap.

Object storage tiers

Cloud platforms offer object storage services in different tiers.

The tier determines how quickly we can read and write the data, and how much it will cost.

From hot to archive, storage gets cheaper and retrieval gets more expensive.



The object storage tier costs

Hot, cold and archive

ONLINE

Hot

For frequently accessed data, always available when we want it. The fastest access, and the most expensive tier of all.

ONLINE

Cold

For data that is not accessed frequently. Cheaper to store than hot, but access is slower and costlier. Platforms often require data to stay for a period, for example 30 days, or charge a penalty.

OFFLINE

Archive

For archival, legal, preservation or compliance data, often kept on tapes. Access can take many hours, and data must be brought to cold or hot first. The cheapest storage, but expensive to read.

Hot, cold and archive compared

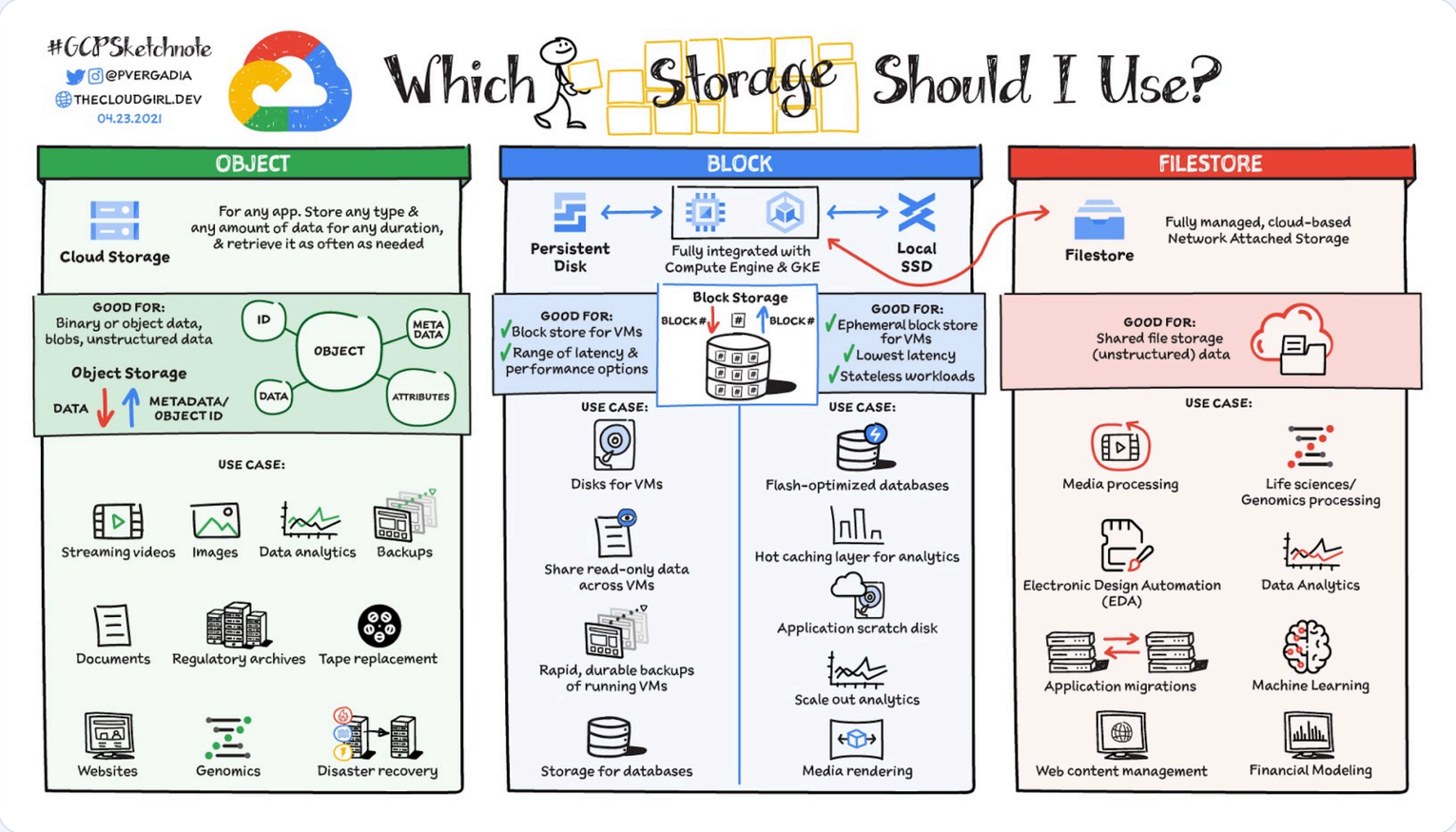
	Hot	Cold	Archive
Access	Frequent	Infrequent	Rare
Type	Online	Online	Offline
Storage cost	Highest	Low	Lowest
Access speed	Fastest	Fast	Slowest
Data retrieval cost	Cheapest	Cheap	Highest

Like cold storage, the archive tier has a minimum storage period.

Block vs file vs object storage

	Block storage	File storage	Object storage
Latency	Low	Medium	High
Scalability	Low	Medium: potentially millions of files, but cannot handle more	High: scales infinitely to petabytes and beyond
Cost	Cheap	Relatively cheaper	Cheaper than block
Accessibility	Local to connected nodes	Local to connected nodes, several other cloud-based services, APIs	Public via APIs
Access	Via volumes by the host OS	SMB/NFS protocols + APIs (cloud)	HTTP/REST protocols

Which storage should I use?



Source: #GCPSketchnote, thecloudgirl.dev

QUIZ

Post-knowledge quiz

Cloud storage requirements, and block, file and object storage.

Next session



01

Cloud networking



02

Content delivery network